

16.4 SIMULTANEITY BIAS

We have mentioned earlier that in simultaneous models the explanatory variables and the error terms are not independent. An important assumption of classical regression model is that there is no covariance between the explanatory variable and the error term.

Let us look into the simple Keynesian model of income determination given in Example 16.2.

$$\text{Consumption function } C_t = \beta_0 + \beta_1 Y_t + u_t \quad 0 < \beta_1 < 1 \quad \dots (16.4)$$

$$\text{Income identity: } Y_t = C_t + I_t \dots (16.5)$$

Suppose we want to estimate the parameters of the consumption (16.4) without considering the income identity (16.5). The standard assumptions under classical regression model for equation (16.4) will be

$$E(u_t) = 0,$$

$$E(u_t)^2 = \sigma^2,$$

$$E(u_t u_{t+i}) = 0 \text{ for } i \neq 0$$

and

$$\text{cov}(Y_t u_t) = 0$$

Note that in this case we take C as the dependent variable and Y as the independent variable. The assumption $\text{cov}(Y_t u_t) = 0$ gets violated in this case, as Y is not independent of u . Let us look into the consequences.

16.4.1 Endogeneity Problem

In the case of the consumption function (16.4) we will first show that Y_t and u_t are correlated and then prove that OLS estimator $\widehat{\beta}_1$ is biased and inconsistent.

To prove that Y_t and u_t are correlated, we substitute (16.4) into (16.5) to obtain the following equation.

$$Y_t = (C_t + I_t) = (\beta_0 + \beta_1 Y_t + u_t) + I_t$$

or,

$$Y_t - \beta_1 Y_t = \beta_0 + u_t + I_t$$

or,

$$(1 - \beta_1)Y_t = \beta_0 + u_t + I_t$$

Therefore,

$$Y_t = \frac{\beta_0}{1-\beta_1} + \frac{1}{1-\beta_1}I_t + \frac{1}{1-\beta_1}u_t \quad \dots(16.14)$$

$$E(Y_t) = \frac{\beta_0}{1-\beta_1} + \frac{1}{1-\beta_1}I_t + \frac{1}{1-\beta_1}E(u_t)$$

or,

$$E(Y_t) = \frac{\beta_0}{1-\beta_1} + \frac{1}{1-\beta_1}I_t \quad \dots (16.15)$$

Subtracting (16.15) from (16.14) results in

$$[Y_t - E(Y_t)] = \frac{1}{1-\beta_1}u_t \quad \dots(16.16)$$

We have assumed that $E(u_t) = 0$. Therefore,

$$u_t = [u_t - E(u_t)] \quad \dots(16.17)$$

The covariance between Y_t and u_t is given by

$$\begin{aligned} cov(Y_t u_t) &= E[y_t u_t] \\ &= E[Y_t - E(Y_t)][u_t - E(u_t)] \end{aligned}$$

Since $[Y_t - E(Y_t)] = \frac{1}{1-\beta_1}u_t$ from (16.16), we have

$$\begin{aligned} cov(Y_t u_t) &= E\left[\frac{1}{1-\beta_1}u_t u_t\right] \\ &= \frac{E(u_t)^2}{1-\beta_1} = \frac{\sigma^2}{1-\beta_1} \quad \dots(16.18) \end{aligned}$$

Since σ^2 is positive by definition, the covariance between Y and u given in (16.18) is bound to be different from zero. As a result, the explanatory variable Y_t and the error term u_t in equation (16.4) are correlated, which violates the assumption of the classical linear regression model that the disturbances are independent of (or not uncorrelated with) the explanatory variables.

In simple language, C determines Y (equation (16.4)), and Y determines C (equation (16.5)). Therefore, we must estimate both the equations together.

The OLS estimator is biased because of correlation between Y_t and u_t . we proceed as follows:

The OLS estimator of β_1 is given by

$$\hat{\beta}_1 = \frac{\sum y_t c_t}{\sum y_t^2} \quad \dots (16.19)$$

In deviation form, equation (16.4) can be expressed as

$$c_t = \beta_1 y_t + u_t$$

By substituting the above equation in (16.19) we obtain

$$\hat{\beta}_1 = \frac{\sum y_t c_t}{\sum y_t^2} = \frac{\sum y_t (\beta_1 y_t + u_t)}{\sum y_t^2} = \beta_1 + \frac{\sum y_t u_t}{\sum y_t^2} \quad \dots (16.20)$$

$$E(\hat{\beta}_1) = \beta_1 + \frac{\sum E(y_t u_t)}{\sum y_t^2}$$

or,

$$E(\hat{\beta}_1) - \beta_1 = \frac{\sum E(y_t u_t)}{\sum y_t^2} \quad \dots (16.21)$$

Equation (16.21) gives the extent of simultaneity bias in the present case. As the expectation-operator is a linear operator, we cannot evaluate $E \left[\frac{\sum y_t u_t}{\sum y_t^2} \right]$, since $E\left(\frac{A}{B}\right) \neq \frac{E(A)}{E(B)}$. But intuitively it is clear that $\hat{\beta}_1$ will be an unbiased estimator of β_1 only if $\left[\frac{\sum y_t u_t}{\sum y_t^2} \right] = 0$.

16.4.3 The OLS Estimator is not Consistent

Though we have shown in (16.18) that the covariance between Y and u is not zero, it does not mean $\text{cov}(y_t u_t) \neq 0$, since $\text{cov}(y_t u_t)$ is a population concept. It is not exactly equal to $\sum y_t u_t$ which is a sample measure, although as the sample size increases indefinitely the latter will tend toward the former. But if the sample size increases indefinitely, then we can resort to the concept of consistent estimator and thus find out what happens to β_1 as n , the sample size, increases indefinitely.

Now an estimator is said to be consistent if its **probability limit**, or **plim** for short, (see Unit 3 for definition) is equal to its true (population) value. Therefore, to show that $\hat{\beta}_1$ of (16.20) is inconsistent; we must show that its *plim* is not equal to the true β_1 .

Applying the rules of probability limit (*plim*) to (16.20), we obtain¹

$$\begin{aligned}\text{plim}(\hat{\beta}_1) &= \text{plim}(\beta_1) + \text{plim}\left(\frac{\sum y_t u_t}{\sum y_t^2}\right) \\ &= \text{plim}(\beta_1) + \text{plim}\left(\frac{\sum y_t u_t / n}{\sum y_t^2 / n}\right) \\ &= \beta_1 + \frac{\text{plim}(\sum y_t u_t / n)}{\text{plim}(\sum y_t^2 / n)} \quad \dots(16.22)\end{aligned}$$

In the second step of equation (16.22) we have divided $\sum y_t u_t$ and $\sum y_t^2$ by the sample size, n , so that the quantities in the parentheses are now the sample covariance between Y and u , and the sample variance of Y respectively.

In other words, (16.22) states that the *plim* of $\hat{\beta}_1$ is equal to true β_1 plus the ratio of two quantities: (i) the *plim* of the sample covariance between Y and u , and (ii) the *plim* of the sample variance of Y . Now, as the sample size n increases indefinitely, we expect the sample covariance between Y and u to approximate the population covariance $E[Y_t - E(Y_t)][u_t - E(u_t)]$. From (16.18) we know that

$$\text{cov}(Y_t u_t) = [\sigma^2 / (1 - \beta_1)]$$

Similarly, as n tends to infinity, the sample variance of Y will approximate its population variance, say σ_y^2 . Therefore, equation (16.22) may be written as

$$\begin{aligned}p \lim(\hat{\beta}_1) &= \beta_1 + \frac{\sigma^2 / (1 - \beta_1)}{\sigma_y^2} \\ &= \beta_1 + \frac{1}{(1 - \beta_1)} \left(\frac{\sigma^2}{\sigma_y^2} \right) \quad \dots(16.23)\end{aligned}$$

It is assumed that $0 < \beta_1 < 1$. Further, σ^2 and σ_y^2 are both positive. Thus, equation (16.23) implies that $\text{plim}(\hat{\beta}_1)$ will always be greater than β_1 ; that is, $\hat{\beta}_1$ will overestimate the true parameter², β_1 . In other words, $\hat{\beta}_1$ is a biased estimator, and the bias will not disappear; no matter how large the sample size is.

16.4.2 Solution to Simultaneity Bias

In the preceding section we learnt that if we apply OLS to an equation belonging to a system of simultaneous equations, we will not get unbiased and consistent estimators. To obtain unbiased and consistent estimators, we apply certain other methods of estimation. There are several methods for this purpose. The most common methods are:

¹As mentioned in Unit 1, the *plim* of a constant (for example, β_1) is the same constant and the *plim* of $(A/B) = \text{plim}(A)/\text{plim}(B)$. Note, however, that $E(A/B) \neq E(A)/E(B)$.

²In general, however, the direction of the bias will depend on the structure of the model and the parameters.

- (i) Reduced form method, or indirect least squares (ILS)
- (ii) Instrumental variables (IV)
- (iii) Two-stage least squares (2SLS)
- (iv) Limited information maximum likelihood (LIML)
- (v) Three stage least squares (3SLS)
- (vi) Full information maximum likelihood (FIML)

Some of these methods will be discussed in the course 'MECE 102: Advanced Econometric Methods'.

16.5 STRUCTURAL FORM AND REDUCED FORM

We presented the simultaneous equations system in (16.13). These equations are known as the 'structural equations' or 'behavioral equations' as they represent the structure (of an economic model) or the behavior of an economic agent (e.g., consumer or producer). The β 's and γ 's in (16.13) are known as the structural parameters. Recall that in the structural equations both endogenous and exogenous variables are on the right-hand side of each equation.

By solving for the M endogenous variables from the structural equations we can derive the reduced-form equations and the associated reduced-form coefficients. A reduced-form equation is one where each equation contains one endogenous variable and that endogenous variable is expressed solely in terms of the predetermined variables and the stochastic disturbances. To illustrate, let us consider the Keynesian model of income determination given at Example 16.2 again:

$$\text{Consumption function: } C_t = \beta_0 + \beta_1 Y_t + u_t \quad \dots(16.24)$$

$$\text{where } 0 < \beta_1 < 1$$

$$\text{Income identity: } Y_t = C_t + I_t \quad \dots(16.25)$$

In the above model C (consumption) and Y (income) are endogenous variables and I (investment expenditure) is an exogenous variable. While the first equation is stochastic in nature, the second one is an identity. As usual, the MPC (β_1) is assumed to remain between 0 and 1. Equations (16.24) and (16.25) are called the 'structural form' of the model. Here, β_0 and β_1 are structural parameters of the model.

If we substitute (16.24) in (16.25), we obtain the following:

$$Y_t = C_t + I_t = [\beta_0 + \beta_1 Y_t + u_t] + I_t$$

By re-arranging terms in the above equation, we obtain

$$\begin{aligned} Y_t - \beta_1 Y_t &= \beta_0 + I_t + u_t \\ \text{or,} \\ (1 - \beta_1) Y_t &= \beta_0 + I_t + u_t \end{aligned}$$

or,

$$Y_t = \frac{\beta_0}{(1 - \beta_1)} + \frac{1}{(1 - \beta_1)} I_t + \frac{1}{(1 - \beta_1)} u_t$$

The above can be represented as

$$Y_t = \Pi_0 + \Pi_1 I_t + w_t \quad \dots(16.26)$$

where $\Pi_0 = \frac{\beta_0}{1 - \beta_1}$

$$\Pi_1 = \frac{1}{1 - \beta_1}$$

$$w_t = \frac{u_t}{1 - \beta_1}$$

Equation (16.26) is a 'reduced form' equation. It expresses the endogenous variable Y solely as a function of the exogenous (or pre-determined) variable and the stochastic disturbance term. Π_0 and Π_1 are 'reduced-form parameters'. Notice that these reduced-form parameters are nonlinear combinations of the structural parameters.

Substituting the value of Y_t from (16.26) into C of (16.24), we obtain another reduced-form equation:

$$C_t = \Pi_2 + \Pi_3 I_t + w_t \quad \dots(16.27)$$

where $\Pi_0 = \frac{\beta_0}{1 - \beta_1}, \Pi_3 = \frac{\beta_0}{1 - \beta_1}, \text{ and } w_t = \frac{u_t}{1 - \beta_1}$

The reduced-form coefficients, such as Π_1 and Π_2 and Π_3 , are also known as impact, or short-run multipliers, because they measure the immediate impact on the endogenous variable of a unit change in the value of the exogenous variable. If in the preceding Keynesian model the investment expenditure is increased by, say, Re.1 and if the MPC is assumed to be 0.8, then from (16.25) we obtain $\Pi_1 = 5$. This result means that increasing the investment by Re.1 will immediately (i.e., in the current time period) lead to an increase in income of Rs. 5, that is, a fivefold increase. Similarly, under the assumed conditions, equation (16.25) shows that $\Pi_3 = 4$, meaning that Re. 1 increase in investment expenditure will lead immediately to an increase of Rs. 4 in consumption expenditure.

In the context of econometric models, equations such as (16.25) or $Q_{dt} = Q_{st}$ (Quantity demanded is equal to quantity supplied) are known as equilibrium condition. On the other hand, identity states that aggregate income Y must be equal to aggregate expenditure (i.e., consumption expenditure plus investment

expenditure). When equilibrium is achieved, the endogenous variables assume their equilibrium values³.

16.6 CONCEPT OF IDENTIFICATION

In the previous section we presented the reduced form of an equation. In the reduced form an endogenous variable is explained in terms of the pre-determined variables in the simultaneous equations system and the stochastic error term. Thus, we can apply OLS to the reduced form and obtain estimators of the reduced form parameters.

Our objective, however, is to find out the structural form parameters. The concept of identification is used to denote the possibility (or impossibility) of deducing the values of the structural parameters from the reduced form parameters.

The identification issue arises prior to estimation of the model. If more than one theory is consistent with the same data (we illustrate this in Example 16.4 below), then they are said to be observationally equivalent, and we cannot distinguish them. In such cases the structure of the econometric model is said to be unidentified and there is a need to modify the model.

Identification is a problem of model formulation, rather than of model estimation or model appraisal. A model is said to be 'identified' if it is in a unique statistical form, where we can obtain unique estimator for each of its structural parameters. If a model is 'not identified', then we cannot say exactly what relationship we are estimating.

Example 16.4

Identification Problem in the Demand-Supply Model

To measure the coefficients of the demand equation, normally the published time series reporting the quantity bought of the commodity is used. However, the quantity bought is identical with the quantity sold at the given price. Market data register points of interaction of equilibrium supply and demand at the price prevailing in the market at a certain point of time. A sample of time-series observations shows simultaneously the quantity demanded, and the quantity supplied, at the prevailing market price. That is, it only shows the points of interactions of demand and supply. If we use these data for estimation, we measure the coefficients of a function of the form $Q = f(p)$. This equation may be either the demand function or the supply function. How can we be sure whether this equation represents demand function or supply function? If person A is interested in estimation of the demand function, then (s)he can use the market data. Similarly, if person B is interested in estimation of the supply equation, (s)he will also be using the same market data! Obviously, we need some criteria, which will enable us to verify that the estimated coefficients belong to a particular relationship. Such criteria are known as 'identification conditions' of a function. We will explain these criteria in Section 16.6.

³ For details see Jan Kmenta (1997), *Elements of Econometrics*, 2nd Edition, Macmillan, New York, pp. 723-731.

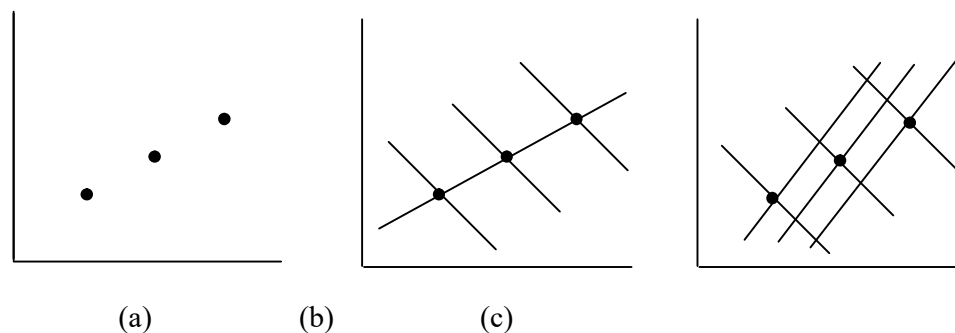


Fig. 16.2: Supply and Demand Functions

In Fig. 16.2 we plot market data for three observations. Fig. 16.2(a) shows three data points. We have no knowledge of the conditions of supply and demand beyond our consideration that the data represents the equilibrium points, that is, the equality between supply and demand. In Fig. 16.2(b) we try to introduce supply and demand functions for these three observations. Notice that the supply curve is stable in Fig. 16.1(b) whereas the demand curve is shifting. Thus, the data represents points on the supply equation under the assumption that the supply curve is stable and the demand curve shifts. In Fig. 16.2(c), on the other hand, we show that both the demand curve and the supply curve are shifting randomly. Thus, if we wish to estimate $Q = f(p)$ from the same data set, then we estimate neither the demand curve nor the supply curve; rather we end up estimating the mongrel equation⁴.

16.6.1 Paradox of Identification

Discussion in the preceding section suggests that if we want to measure a given function belonging to a simultaneous-equations model, the function must be fairly stable over the sample period, that is, it must shift within a smaller range as compared with other relationships of the same model. In Example 16.4 we have shown that we can measure the supply function when it is fairly stable and the demand function is shifting. Similarly, we can measure the demand function if it is fairly stable while the supply function shows adequate variability.

It can happen when the factor causing a shift in a function is absent in another function. In other words, to identify the supply function (see Fig. 16.2(b)), some factors absent from it but included in the demand function. So, to identify an equation in a simultaneous equations model, we have to look for the variables absent from it (but present in other equations of the model). This is called the paradox of identification. In identification problem, the relevant variables are the ones that are absent from the equation!

Example 16.5

Let us consider the supply-demand equation system. We write the system in equilibrium form (that is, with $q_s = q_d = q$) as

$$q_s = \alpha_1 p + \beta_1 z_1 + u_1 \quad \dots(16.28)$$

and

⁴ A combination of two and more equations is called *amongrel* equation.

$$q_d = \alpha_2 p + u_2 \quad \dots (16.29)$$

To be more precise, let q be per capita rice consumption at the country level, let p be the average price per kg of rice in the county, and let z_1 be the price of fertilizer. While q and p are assumed to be endogenous, z_1 is assumed to be exogenous. It means that (16.28) must be the supply function, as the price of fertilizer would shift supply ($\beta_1 < 0$) but not demand. The demand function contains no exogenous variables.

Intuitively, we can say that the demand equation (16.29) is identified. We have an observed variable z_1 that shifts the supply equation; but it does not affect the demand equation. Thus, the demand function can be traced by taking different values of z_1 .

The supply equation (16.28), on the other hand, cannot be traced out because there is no exogenous observed factor that leads to a shift in the demand curve. Only the stochastic error term (which is not observable) leads to a shift in the demand equation. May be in the presence of certain factors (such as preference of people towards wheator increase in per capita income) we can identify the supply equation.

To summarize: in the system of (16.28) and (16.29), it is the presence of an exogenous variable (z_1) in the supply equation that allows us to estimate the demand equation. Further, we have to check the identification status of separately for each equation in the model.

Let us consider a simultaneous equations model with two equations.

$$y_1 = \alpha_1 + \beta_1 y_2 + z_1 \beta_2 + u_1 \quad \dots (16.30)$$

and

$$y_2 = \alpha_2 + \beta_3 y_1 + z_2 \beta_4 + u_2 \quad \dots (16.31)$$

where y_1 and y_2 are the endogenous variables, and u_1 and u_2 are the error terms. The intercept in the first equation is α_1 , and the intercept in the second equation is α_2 . The fact that z_1 and z_2 are different exogenous variables means that we have imposed exclusion restrictions on the model. In other words, we assume that certain exogenous variables do not appear in the first equation while some exogenous variables do not appear in the second equation. The key question is: Under what conditions can we estimate the parameters in, say, (16.31)? This is the identification issue.

16.6.2 Identification Status of an Equation

As mentioned earlier, identification refers to the possibility of finding out estimators of all the parameters of an equation. Thus, we have to move equation-wise. There are two possibilities: whether the equation under consideration is identified or not identified.

1. Equation is under-identified

If the structure of the equation is not unique or there is not sufficient information to uniquely distinguish the equation, then the equation is under-identified.

2. Equation is identified

- (a) Exactly identified: If there is sufficient information available to distinguish the equation then it is called just identified or exactly identified equation
- (b) Over-identified: If there is more than sufficient information regarding the equation then the equation is called over-identified.

Identification is closely related to the estimation of the model.

- (1) If an equation is identified, its coefficient can, in general, be statistically estimated. In particular:
 - (a) If the equation is exactly identified, the appropriate method to be used for its estimation is the method of indirect least squares (ILS).
 - (b) If the equation is over-identified, indirect least squares cannot be applied, because it will not yield unique estimates of the structural parameters. There are various methods which can be used in this case, for example two-stage least squares (2SLS), or maximum likelihood methods. These methods will be developed in Unit 15.

Identification may be established either by the examination of the specification of the structural model, or by the examination of the reduced form of the model. Reduced form approach is conceptually confusing and computationally more difficult than the structural model approach, because it requires the derivation of the reduced form first and then examination of the values of the determinant formed from some of the reduced form coefficients. The structural form approach is simpler and more useful. In this unit we will only discuss the structural form approach of identification.

In applying the identification rules, we should ignore the intercept term if it is present in each equation. If intercept term is present in some equations of the model, not in others, then we should treat it like a separate variable (say X_0), that takes the value 1 for presence of the intercept term, and 0 otherwise.

Check Your Progress 2

- 1) Explain the following concepts:

- Structural equations
- Reduced Form Equations
- Simultaneity Bias

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- 2) Why is it called the paradox of identification?

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